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STABILITY FOR AN INVERSE PROBLEM IN POTENTIAL THEORY

HAMID BELLOUT, AVNER FRIEDMAN AND VICTOR ISAKOV

ABSTRACT. Let D be a subdomain of a bounded domain Ω in $\mathbb{R}^n$. The conductivity coefficient of D is a positive constant $k \neq 1$ and the conductivity of $\Omega \setminus D$ is equal to 1. For a given current density g on $\partial\Omega$, we compute the resulting potential u and denote by f the value of u on $\partial\Omega$. The general inverse problem is to estimate the location of D from the known measurements of the voltage f . If D_h is a family of domains for which the Hausdorff distance $d(D, D_h)$ equal to $O(h)$ (h small), then the corresponding measurements f_h are $O(h)$ close to f . This paper is concerned with proving the inverse, that is, $d(D, D_h) \leq \frac{1}{c} \|f_h - f\|$, $c > 0$; the domains D and D_h are assumed to be piecewise smooth. If $n \geq 3$, we assume in proving the above result, that $D_h \supset D$ (or $D_h \subset D$) for all small h . For $n = 2$ this monotonicity condition is dropped, provided g is appropriately chosen. The above stability estimate provides quantitative information on the location of D_h by means of f_h .

1. INTRODUCTION

For any two domains D_1, D_2 in $\mathbb{R}^n$ denote by $d(D_1, D_2)$ the Hausdorff distance between them. Let Ω be a bounded domain in $\mathbb{R}^n$ ($n \geq 2$) and let D and D_h (for any $0 < h < h_0$, h_0 small) be subdomains of Ω with closure in Ω such that

$$c_1 h \leq d(D, D_h) \leq c_2 h,$$

where c_1, c_2 are positive constants. Set

$$a_D = \begin{cases} k & \text{in } D, \\ 1 & \text{in } \Omega \setminus D, \end{cases} \quad a_{D_h} = \begin{cases} k & \text{in } D_h, \\ 1 & \text{in } \Omega \setminus D_h, \end{cases}$$

where k is a positive constant, $k \neq 1$. Consider the Neumann problems

$$(1.1) \quad \begin{aligned} \operatorname{div}(a_D \nabla u_D) &= 0 & \text{in } \Omega, \\ \frac{\partial u_D}{\partial \nu} &= g & \text{on } \partial\Omega, \end{aligned} \quad \int_{\Omega} u_D = 0$$

and

$$(1.2) \quad \begin{aligned} \operatorname{div}(a_{D_h} \nabla u_{D_h}) &= 0 & \text{in } \Omega, \\ \frac{\partial u_{D_h}}{\partial \nu} &= g & \text{on } \partial\Omega, \end{aligned} \quad \int_{\Omega} u_{D_h} = 0,$$

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where g is a given function, satisfying

$$(1.3) \quad g \in L^2(\partial\Omega), \quad g \neq 0, \quad \int_{\partial\Omega} g = 0,$$

and set

$$(1.4) \quad f = u_D|_{\partial\Omega}, \quad f_h = u_{D_h}|_{\partial\Omega}.$$

We are interested in establishing a local stability estimate of the form

$$(1.5) \quad d(D, D_h) \leq C|f - f_h|_{L^1(\Gamma)}$$

where Γ is a nonempty open subset of $\partial\Omega$, and h is sufficiently small. Such an estimate means that the mapping

$$D \rightarrow \mathcal{F}(D) \equiv u_D|_{\Gamma}$$

has nonzero “derivative.”

We shall refer to the case

$$(1.6) \quad D_h \supset D \quad \text{for all } h \text{ (or } D_h \subset D \text{ for all } h)$$

as the *monotone case*. When the assumption (1.6) is dropped, we speak of the *nonmonotone case*.

We shall always assume that ∂D is piecewise smooth, and that ∂D_h has the representation

$$(1.7) \quad \partial D_h: x = f(s) + h\sigma_h(s)\nu(s) \quad \text{a.e.}$$

where $\nu(s)$ is the normal to ∂D , wherever it exists, and $|\sigma_h(s)| \leq C$; s is an $(n-1)$ -dimensional local parameter. Notice that ∂D is given by $x = f(s)$.

Bellout and Friedman [1] established (1.5) in the monotone case, provided ∂D and ∂D_h are in $C^{2,\alpha}$ (uniformly in h); their proof actually requires only $C^{1,1}$ smoothness. An earlier proof of (1.5) for $n=2$, due to Friedman and Gustafsson [5], also required the same smoothness.

For $n=2$ Bellout and Friedman [1] have established (1.5) for the nonmonotone case provided ∂D is analytic and certain finite number of “orthogonality” conditions are satisfied; it is however not easy to verify such conditions even, for instance, if the D_h are translates of D .

In §4 we shall extend the stability result (1.5) of Bellout and Friedman to the monotone case when ∂D is only piecewise $C^{1,1}$; the proof requires some new ideas and technical estimates which are developed in §§2, 3. Our interest in the piecewise $C^{1,1}$ case and in particular in polyhedra stems from a recent uniqueness theorem due to Friedman and Isakov [6]. They proved that if D and D' are any convex polyhedra in Ω such that the solution u_D of (1.1) and the corresponding solution $u_{D'}$ for D' satisfy: $u_D = u_{D'}$ on an open nonempty portion Γ of $\partial\Omega$, then $D = D'$. They needed to assume that either Ω is a half-space or D and D' are not “too close” to $\partial\Omega$. They also established (1.5), but only when Ω is a half-space and under some severe restrictions on D .

In §5 we consider the case $n=2$ and ∂D analytic, but drop the monotonicity assumption (1.6). We establish the stability estimate (1.5) for appropriately chosen function g .

Finally in §6 we extend the results of §5 to the case where D is a convex polygon or, more generally, piecewise analytic.

2. THE BEHAVIOR OF ∇u NEAR A VERTEX OF ∂D

Throughout this paper we assume that Ω is a bounded domain in $\mathbb{R}^n$ ($n \geq 2$) with $C^{1,\alpha}$ boundary.

Let D be a subdomain of Ω with $\overline{D} \subset \Omega$ and let

$$a(x) = \begin{cases} k & \text{if } x \in D, \\ 1 & \text{if } x \in \Omega \setminus D, \end{cases}$$

where k is a positive number $\neq 1$. Consider the diffraction problem

$$(2.1) \quad \operatorname{div}(a \nabla u) = f \quad \text{in } \Omega,$$

$$(2.2) \quad \frac{\partial u}{\partial \nu} = g \quad \text{on } \partial \Omega, \quad \int_{\Omega} u = 0,$$

where g satisfies

$$(2.3) \quad g \in L^2(\partial \Omega), \quad g \neq 0,$$

$$(2.4) \quad \int_{\partial \Omega} g = \int_{\Omega} f.$$

We shall be interested in the behavior of ∇u near a point $x_0 \in \partial D$ where ∂D is not smooth. For simplicity we first consider the case when $n = 2$, $x_0 = 0$ and, for some ball $B_{r_0} \equiv B_{r_0}(x_0)$,

$$(2.5) \quad \begin{aligned} &\partial D \cap \overline{B}_{r_0} \text{ consists of line segments} \\ &l_1 = \{(r, \theta); 0 \leq r \leq r_0, \theta = 0\}, \quad \text{and} \\ &l_2 = \{(r, \theta); 0 \leq r \leq r_0, \theta = \beta\}, \quad 0 < \beta < \pi. \end{aligned}$$

Consider first the case $f = 0$ and set $u^e = u_{\Omega \setminus D}$, $u^i = u_D$.

Lemma 2.1. *The following expansion holds for $0 < r < r_1$ ($r_1 = r_0/2$):*

$$(2.6) \quad \begin{aligned} u^e &= u^e(0) + \sum_{j=1}^{\infty} r^{\gamma_j} (A_j^e \cos \gamma_j \theta + B_j^e \sin \gamma_j \theta), \\ u^i &= u^i(0) + \sum_{j=1}^{\infty} r^{\gamma_j} (A_j^i \cos \gamma_j \theta + B_j^i \sin \gamma_j \theta); \end{aligned}$$

the series are convergent with their first derivatives, absolutely and uniformly for $0 < r \leq r_1$. Here, the sequence γ_j is monotone increasing,

$$(2.7) \quad 0 < c_1 \leq \gamma_j/j \leq c_2 < \infty \quad \text{for all } j,$$

and

$$(2.8) \quad \gamma_1 > \frac{1}{2}.$$

Proof. Denote by S^1 the unit circle and define on S^1

$$a(\theta) = \begin{cases} k & \text{if } 0 \leq \theta \leq \beta, \\ 1 & \text{if } \beta < \theta < 2\pi. \end{cases}$$

Introduce the function spaces $L_a^2(S^1)$, $H_a^1(S^1)$ with norms

$$\|v\|_{L_a^2(S^1)} = \left\{ \int_0^{2\pi} a|v(\theta)|^2 d\theta \right\}^{1/2},$$

$$\|v\|_{H_a^1(S^1)} = \left\{ \int_0^{2\pi} av_\theta^2(\theta) d\theta + \int_0^{2\pi} av^2(\theta) d\theta \right\}^{1/2}.$$

Set

$$(2.9) \quad \mathcal{L}v = \frac{1}{a} \frac{\partial}{\partial \theta} \left(a \frac{\partial}{\partial \theta} v \right);$$

$\mathcal{L}$ is an unbounded, selfadjoint, positive elliptic operator with dense domain in $L_a^2(S^1)$, and $(\mathcal{L} + 1)^{-1}$ is compact. Hence the spectrum of $\mathcal{L}$ consists of positive eigenvalues γ_j^2 ($\gamma_j > 0$). We denote a corresponding (complete) orthonormal sequence by $\{v_j\}$; it is a basis for $L_a^2(S^1)$.

If

$$(2.10) \quad \mathcal{L}v + \gamma_j^2 v = 0,$$

then

$$(2.11) \quad v'' + \gamma_j^2 v = 0 \text{ on } 0 < \theta < \beta \text{ and on } \beta < \theta < 2\pi,$$

so that

$$\begin{aligned} v_j &= v_j^i = M_j^i \cos \gamma_j \theta + N_j^i \sin \gamma_j \theta, & 0 < \theta < \beta, \\ v_j &= v_j^e = M_j^e \cos \gamma_j \theta + N_j^e \sin \gamma_j \theta, & \beta < \theta < 2\pi; \end{aligned}$$

in addition, the diffraction (or transmission) conditions

$$(2.12) \quad \left. \begin{aligned} v_j^i &= v_j^e \\ kv_{j,\theta}^i &= v_{j,\theta}^e \end{aligned} \right\} \text{ at } \theta = 0, \theta = \beta$$

must be satisfied, as well as the condition

$$(2.13) \quad \int_0^{2\pi} v_j^2 d\theta = 1.$$

For every $r \in (0, r_0)$ we have the expansion

$$(2.14) \quad u(r, \theta) = u(0) + \sum_{j=1}^{\infty} h_j(r) v_j(\theta) \text{ in } L^2(S^1)$$

and

$$h_j(r) = \int_0^{2\pi} a u v_j d\theta.$$

The function u satisfies, for $0 < r < r_0$,

$$(2.15) \quad a \left(\frac{1}{r} (ru_r)_r \right) + \frac{1}{r^2} (au_\theta)_\theta = 0.$$

Multiplying (2.15) by $av_j(\theta)$ and integrating over S^1 we get, after using (2.10),

$$(2.16) \quad \frac{1}{r} \left[r \left(\int_{S^1} a u v_j d\theta \right)_r \right] - \frac{\gamma_j^2}{r^2} \int_{S^1} a u v_j = 0.$$

Hence $h_j(r)$ satisfies

$$(2.17) \quad \frac{1}{r}(rh_{j,r})_r - \frac{\gamma_j^2}{r^2}h_j = 0,$$

so that

$$(2.18) \quad h_j(v) = C_j r^{\gamma_j} + D_j r^{-\gamma_j}.$$

Since $u_r \in L^2(B_{r_0})$ we have

$$(2.19) \quad \int_0^{r_0} [(uv_j)_r]^2 r dr \leq M < \infty,$$

i.e.,

$$(2.20) \quad \int_0^{r_0} |h'_j(r)|^2 r dr \leq M.$$

It follows that $D_j = 0$ and, consequently, from (2.14),

$$(2.21) \quad u(r, \theta) = u(0) + \sum C_j r^{\gamma_j} v_j(\theta)$$

and

$$\int_{S^1} au^2(r, \theta) d\theta = u^2(0) + \sum C_j^2 r^{2\gamma_j} \quad (0 < r \leq r_0).$$

Since $u_r \in L_a^2(S^1)$ for $r = r_0$, we actually even have

$$\int_{S^1} au_r^2(r, \theta) d\theta = \sum C_j^2 \gamma_j^2 r^{2(\gamma_j-1)} < \infty$$

for $r = r_0$, so that

$$(2.22) \quad \sum_{j=1}^{\infty} C_j^2 \gamma_j^2 r_0^{2\gamma_j} < \infty.$$

We next estimate the γ_j . We can write

$$v_j^e = \operatorname{Re}\{a_j e^{i\gamma_j \theta}\}, \quad v_j^i = \operatorname{Re}\{b_j e^{i\gamma_j \theta}\}.$$

The refraction conditions at $\theta = 0$ and $\theta = \beta$ then become (for $a = a_j$, $b = b_j$, $\gamma = \gamma_j$)

$$(2.23) \quad a + \bar{a} = be^{i\gamma 2\pi} + \bar{b}e^{-i\gamma 2\pi},$$

$$(2.24) \quad ae^{i\gamma \beta} + \bar{a}e^{-i\gamma \beta} = be^{i\gamma \beta} + \bar{b}e^{-i\gamma \beta}$$

and

$$(2.25) \quad k(a - \bar{a}) = be^{i\gamma 2\pi} - \bar{b}e^{-i\gamma 2\pi},$$

$$(2.26) \quad k(ae^{i\gamma \beta} - \bar{a}e^{-i\gamma \beta}) = be^{i\gamma \beta} - \bar{b}e^{-i\gamma \beta}.$$

Taking k times (2.23) and adding to (2.25), we get

$$(2.27) \quad 2ka = (k+1)be^{i\gamma 2\pi} + (k-1)\bar{b}e^{-i\gamma 2\pi}.$$

Similarly, taking k times (2.24) and adding to (2.26), we get

$$(2.28) \quad 2ka = (k+1)b + (k-1)\bar{b}e^{-2i\gamma \beta}.$$

Comparing (2.27) with (2.28) we find that

$$(2.29) \quad (k+1)b(e^{i\gamma 2\pi} - 1) = (k-1)\bar{b}(e^{-2i\gamma\beta} - e^{-2i\gamma\pi}).$$

We need to consider two cases

Case (i). $e^{2i\gamma\pi} \neq 1$.

Then

$$\frac{b}{\bar{b}} = \frac{k-1}{k+1} \frac{e^{-2i\gamma\beta} - e^{-2i\gamma\pi}}{e^{2i\gamma\pi} - 1}.$$

Since $|b/\bar{b}| = 1$, we conclude that

$$(2.30) \quad \frac{|e^{-2i\gamma\beta} - e^{-2i\gamma\pi}|}{|e^{2i\gamma\pi} - 1|} = \left| \frac{k+1}{k-1} \right| \equiv A > 1,$$

or

$$(2.31) \quad \sin \gamma(\pi - \beta) = A \sin \gamma\pi$$

and it is easy to see that this equation has an infinite sequence of solutions γ_j satisfying (2.7). We claim that the smallest one, γ_1 , satisfies $\gamma_1 > \frac{1}{2}$. Indeed, if $\gamma_1 \leq \frac{1}{2}$ then $2\pi\gamma_1 < \pi$ and $0 < 2\gamma_1\beta < 2\pi\gamma_1 < \pi$. But then

$$|e^{2i\gamma_1\pi} - e^{2i\gamma_1\beta}| < |e^{2i\gamma_1\pi} - 1|,$$

a contradiction to (2.30).

Case (ii). $e^{2i\gamma\pi} = 1$.

Then $\gamma = \gamma_j = n$ for some integer n , and from (2.29) we see that $\beta\gamma/\pi$ is also an integer; consequently

$$(2.32) \quad \beta = \frac{q}{m}\pi, \quad q \text{ and } m \text{ are relatively prime positive integers.}$$

We easily see that all the additional solutions γ , in this case, are multiples of m . Thus the asserted expansion (2.14) still holds, but one has to include the additional sequence of multiples of m into the sequence of the γ_j 's.

Finally, using (2.22) it is easily seen that the series expansion of $u(r, \theta)$ and its gradient are absolutely uniformly convergent for $0 < r \leq r_0/2$.

We shall now extend Lemma 2.1 to the case $f \not\equiv 0$, assuming that $f \in L^{4/3}(\Omega)$.

Set

$$f_j(r) = \int_0^{2\pi} f(r, \theta) v_j(\theta) d\theta.$$

Then formally

$$(2.33) \quad u(r, \theta) = u(0) + \sum C_j r^{\gamma_j} v_j(\theta) + \sum e_j(r) v_j(\theta)$$

where

$$(2.34) \quad e_j(r) = \frac{r^{\gamma_j}}{2\gamma_j} \int_{r_0/2}^r f_j(s) s^{1-\gamma_j} ds - \frac{r^{-\gamma_j}}{2\gamma_j} \int_0^r f_j(s) s^{1+\gamma_j} ds$$

is a solution of

$$\frac{1}{r}(rh')' - \frac{\gamma_j^2}{r^2}h = f_j(r);$$

the fact that $D_j = 0$ follows by using (2.19) as before, noting that $e_j = O(r^2)$, $e'_j = O(r)$. Observe that the first integral on the right-hand side of (2.34)

is from $r_0/2$ to r (the integral from 0 to r will not converge if $\gamma_j \geq 2$). We have,

(2.35)

$$\begin{aligned} \left| \int_0^r f_j(s) s^{1+\gamma_j} ds \right| &= \left| \int_0^r \int_0^{2\pi} f(s, \theta) v_j(\theta) s^{1/p} s^{1+\gamma_j} s^{1-1/p} d\theta ds \right| \\ &\leq \left(\int_0^r \int_0^{2\pi} |f|^p s d\theta ds \right)^{1/p} \left(\int_0^r \int_0^{2\pi} s^{1+(1+\gamma_j)q} |v_j(\theta)|^q d\theta ds \right)^{1/q} \quad \left(q = \frac{p}{p-1} \right) \\ &\leq \frac{C r^{1+\gamma_j+1/q}}{\gamma_j^{1/q}} \left(\int |v_j(\theta)|^q d\theta \right)^{1/q}, \end{aligned}$$

if $p = \frac{4}{3}$, $q = 4$ (since $f \in L^{4/3}$). Noting that by Sobolev's imbedding [4, p. 27],

$$|v_j|_{L^3} \leq C(|v'_j|_{L^2})^a (|v_j|_{L^2})^{1-a}, \quad a = \frac{1}{4},$$

and

$$|v_j|_{L^2} = 1, \quad |v'_j|_{L^2} \leq C \gamma_j |v_j|_{L^2} = C \gamma_j,$$

we get

$$|v_j|_{L^3} \leq C \gamma_j^{1/4}.$$

Substituting this into (2.35), we get

$$\left| \int_0^r f_j(s) s^{1+\gamma_j} ds \right| \leq \frac{C r^{1+\gamma_j+1/4}}{\gamma_j^{1/4}}.$$

A similar estimate holds for the second integral in (2.34). Hence

$$(2.36) \quad \sum |e_j(r) v_j(\theta)| \leq \sum \frac{C r^2}{\gamma_j^{1+1/4}} \leq C r^2,$$

by (2.7).

From (2.33), (2.36) we deduce that the series

$$(2.37) \quad \sum_j C_j r^{\gamma_j} v_j(\theta)$$

is convergent in $L^2(S^1)$ and therefore

$$\sum C_j^2 r^{2\gamma_j} < \infty, \quad 0 < r < r_0.$$

This implies the absolute uniform convergence of the series (2.37) for $0 < r \leq r_0/2$; in particular,

$$(2.38) \quad |u(r, \theta) - u(0)| \leq C r^{\gamma_1}, \quad \gamma_1 > \frac{1}{2}.$$

We now consider the function

$$v_\lambda(x) = u(\lambda x) - u(0)$$

for λ small and x in $B_* = \{\frac{1}{4} < |x| < 4\}$. Let $B_0 = \{\frac{1}{2} < |x| < \frac{7}{2}\}$. Clearly

$$\operatorname{div}(a \nabla v_\lambda) = \lambda^2 f(\lambda x), \quad |v_\lambda| \leq C \lambda^{\gamma_1}.$$

Let l_λ be any line in $\tilde{B} = \{1 < |x| < 3\}$ with endpoints on $\partial\tilde{B}$. Then, by the trace imbedding (of $H^{1/2}(B) \rightarrow L^2(l_\lambda)$), Sobolev's imbedding [7, p. 27] and L^p elliptic estimates,

$$\left\{ \int_{l_\lambda} |\nabla v_\lambda|^2 dx \right\}^{1/2} \leq C |v_\lambda|_{W^{2,4/3}(B_0)} \leq C \int_{B_*} |\lambda^2 f(\lambda x)|^{4/3} dx + C_1 \lambda^{\gamma_1}.$$

Making the substitution $\lambda x = y$ we find that

$$\int_l |\nabla u|^2 \leq C \left(\int_{\lambda/4 < |x| < \lambda} |f|^{4/3} \right)^{3/2} + C \lambda^{\gamma_1 - 1/2},$$

where l is the image of l_λ ; l is any interval connecting a point on $\{r = 1/\lambda\}$ to a point on $\{r = 3/\lambda\}$. By varying λ , taking for instance $\lambda = 3^{-j}$, we deduce that

$$(2.39) \quad \int_l |\nabla u|^2 \leq C \varepsilon(|l|), \quad \varepsilon(t) \downarrow 0 \text{ if } t \downarrow 0$$

where l is an interval in $\{r < \varepsilon_0\}$ and $|l|$ = length of l .

If D is a polygon, then by applying the above estimate near each vertex of ∂D we arrive at the following result:

Lemma 2.2. *Suppose D is a polygon and $f \in L^{4/3}(\Omega)$. Then the solution of the refraction problem (2.1), (2.2) satisfies:*

For any family of intervals which are the intersection of straight lines parallel to one of the sides of D and Ω_0 , a compact subset of Ω ,

$$(2.40) \quad \|\nabla u\|_{L^2(l)} \leq C \|f\|_{L^{4/3}(\Omega)}$$

where C is a constant depending only on D , Ω_0 and g ; furthermore, for any vertex a of ∂D ,

$$(2.41) \quad \|\nabla u\|_{L^2(l \cap B(a, r))} \leq C \varepsilon(|l|), \quad \varepsilon(t) \downarrow 0 \text{ if } t \downarrow 0.$$

Extension of this result to piecewise smooth domain in any number of dimension will be discussed in §4.

3. AN AUXILIARY ESTIMATE

Let D be a polygon in $\mathbb{R}^2$ with edges $\Gamma_1, \Gamma_2, \dots, \Gamma_N$ and vertices $s_1, s_2, \dots, s_N$ such that $s_j = \overline{\Gamma_j} \cap \overline{\Gamma_{j+1}}$, $\Gamma_{N+1} = \Gamma_1$. Let D_h ($0 < h \leq h_0$) be a family of polygons with edges $\Gamma_1(h), \Gamma_2(h), \dots, \Gamma_N(h)$ and vertices $s_1(h), \dots, s_N(h)$ such that $s_j(h) = \overline{\Gamma_j(h)} \cap \overline{\Gamma_{j+1}(h)}$, $\Gamma_{N+1}(h) = \Gamma_1(h)$. We assume that

$$(3.1) \quad \begin{aligned} D &\subset D_h, & \overline{D}_h &\subset \Omega \quad \text{for } 0 < h \leq h_0, \\ c_1 h &\leq \sum_{j=1}^N |s_j(h) - s_j| \leq c_2 h & (0 < c_1 < c_2 < \infty). \end{aligned}$$

Set

$$a = \begin{cases} k & \text{in } D, \\ 1 & \text{in } \Omega \setminus D, \end{cases} \quad a_h = \begin{cases} k & \text{in } D_h \\ 1 & \text{in } \Omega \setminus D_h \end{cases} \quad (k > 0, k \neq 1),$$

and consider the diffraction problems

$$(3.2) \quad \operatorname{div}(a \nabla u) = 0 \quad \text{in } \Omega,$$

$$(3.3) \quad \frac{\partial u}{\partial \nu} = g \quad \text{on } \partial \Omega, \quad \int_{\Omega} u = 0$$

and

$$(3.4) \quad \operatorname{div}(a_h \nabla u_h) = 0 \quad \text{in } \Omega,$$

$$(3.5) \quad \frac{\partial u_h}{\partial \nu} = g \quad \text{on } \partial \Omega, \quad \int_{\Omega} u_h = 0$$

where g satisfies

$$(3.6) \quad g \in L^2(\Omega), \quad g \neq 0, \quad \int_{\partial \Omega} g = 0.$$

We are interested in estimating the “quotient difference”

$$(3.7) \quad U_h = \frac{u_h - u}{h}.$$

Lemma 3.1. *For any $0 < \varepsilon < 2$ there is a constant C such that*

$$(3.8) \quad \int_{\Omega} |U_h|^{2+\varepsilon} \leq C \quad \forall 0 < h \leq h_0.$$

For $\varepsilon = 0$ and $\partial D \in C^{1,1}$ this was proven by Bellout and Friedman [1].

Proof. Multiplying the difference of the equations (3.2), (3.4) by a function v in $H^1(\Omega)$ and integrating over Ω , we easily get

$$(3.9) \quad \int_{\Omega} a_h \nabla U_h \cdot \nabla v + \frac{k-1}{h} \int_{D_h \setminus D} \nabla u \cdot \nabla v = 0 \quad \forall v \in H^1(\Omega).$$

We introduce the solution w_h to the diffraction problem

$$(3.10) \quad \operatorname{div}(a_h \nabla w_h) = U_h \quad \text{in } \Omega,$$

$$(3.11) \quad \frac{\partial w_h}{\partial \nu} = 0 \quad \text{on } \partial \Omega, \quad \int_{\Omega} w_h = 0;$$

since $\int_{\Omega} U_h = 0$, this problem does in fact have a unique solution.

Multiplying (3.10) by U_h and integrating over Ω , we get

$$(3.12) \quad - \int_{\Omega} a_h \nabla w_h \cdot \nabla U_h = \int_{\Omega} U_h^2.$$

Substituting $v = w_h$ in (3.9) and adding the result to (3.12), we find that

$$\frac{k-1}{h} \int_{D_h \setminus D} \nabla u \cdot \nabla w_h = \int_{\Omega} U_h^2,$$

so that, by Cauchy's inequality,

$$(3.13) \quad \int_{\Omega} U_h^2 \leq \frac{|k-1|}{h} \left\{ \int_{D_h \setminus D} |\nabla u|^2 \right\}^{1/2} \left\{ \int_{D_h \setminus D} |\nabla w_h|^2 \right\}^{1/2}.$$

By extending Γ_j at s_{j+1} as a line segment until it meets ∂D_h , for $1 \leq j \leq N$, we get a “triangulation” of $D_h \setminus D$ into N quadrangles $Q_j(h)$, each bounded by the extended Γ_{j-1} , Γ_j and portions of $\Gamma_j(h)$, $\Gamma_{j+1}(h)$. Each $Q_j(h)$ can be traced by a family of intervals $l_j(\lambda, h)$ parallel to Γ_j at distance λ from Γ , where λ varies in some interval $0 \leq \lambda \leq h_j$, $h_j \leq Ch$. Hence

$$\begin{aligned} \int_{D_h \setminus D} |\nabla u|^2 &\leq \sum_{j=1}^N \int_{Q_j(h)} |\nabla u|^2 = \sum_{j=1}^N \int_0^{h_j} d\lambda \int_{l_j(\lambda, h)} |\nabla u|^2 \\ &\leq \sum_{j=1}^N \int_0^{h_j} C d\lambda, \quad \text{by Lemma 2.2.} \end{aligned}$$

It follows that

$$(3.14) \quad \int_{D_h \setminus D} |\nabla u|^2 \leq Ch.$$

Similarly

$$\int_{D_h \setminus D} |\nabla w_h|^2 \leq \sum_{j=1}^N \int_0^{h_j} d\lambda \int_{l_j(\lambda, h)} |\nabla w_h|^2,$$

and

$$\int_{l_j(\lambda, h)} |\nabla w_h|^2 \leq C \int_{\Omega} |U_h|^2,$$

by Lemma 2.2 applied to $w_h / \{\int_{\Omega} |U_h|^2\}^{1/2}$; hence

$$(3.15) \quad \int_{D_h \setminus D} |\nabla w_h|^2 \leq Ch \int_{\Omega} U_h^2.$$

Substituting the estimates (3.14), (3.15) into the right-hand side of (3.13), we conclude that

$$\int_{\Omega} U_h^2 \leq C \left\{ \int_{\Omega} U_h^2 \right\}^{1/2},$$

i.e.,

$$(3.16) \quad \int_{\Omega} U_h^2 \leq C.$$

Having proved (3.8) for $\varepsilon = 0$ we proceed to prove it for ε positive and small. For this purpose we introduce another auxiliary function w_h defined as the solution to

$$(3.17) \quad \operatorname{div}(a_h \nabla w_h) = |U_h|^{\varepsilon} U_h - A \equiv F_{\varepsilon} \quad \text{in } \Omega$$

with the same conditions (3.11) as before; the constant A is chosen so that F_{ε} satisfies the compatibility condition $\int_{\Omega} F_{\varepsilon} = 0$, that is

$$A = \frac{1}{|\Omega|} \int_{\Omega} |U_h|^{\varepsilon} U_h.$$

From (3.16) it follows that

$$\int_{\Omega} |F_{\varepsilon}|^{4/3} \leq C,$$

if

$$\frac{4}{3} = \frac{2}{1+\varepsilon}, \quad \text{i.e., if } \varepsilon = \frac{1}{2}.$$

We can then apply Lemma 2.2 and deduce that

$$\int_{l_j(\lambda, h)} |\nabla w_h|^2 \leq C,$$

for any line $l_j(\lambda, h)$ and, consequently,

$$(3.18) \quad \int_{D_h \setminus D} |\nabla w_h|^2 \leq Ch.$$

Next we multiply (3.17) by U_h and integrate over Ω . Since $\int_{\Omega} U_h = 0$, we obtain

$$(3.19) \quad - \int_{\Omega} a_h \nabla w_h \cdot \nabla U_h = \int_{\Omega} |U_h|^{2+\varepsilon}.$$

Substituting $v = w_h$ in (3.9) and adding to (3.19), we find that

$$(3.20) \quad \int_{\Omega} |U_h|^{2+\varepsilon} = \frac{k-1}{h} \int_{D_h \setminus D} \nabla u \cdot \nabla w_h,$$

and using the estimates (3.14), (3.18), we get

$$(3.21) \quad \int_{\Omega} |U_h|^{2+\varepsilon} \leq C$$

where C is a constant independent of h , or (since $\varepsilon = 1/2$)

$$\int_{\Omega} |U_h|^{2+1/2} \leq C,$$

which is an improvement of (3.16). More generally, assuming that (3.8) holds for $\varepsilon = \varepsilon_m$ the above proof shows that (3.8) will then hold for $\varepsilon = \varepsilon_{m+1}$ where

$$\frac{4}{3} = \frac{2 + \varepsilon_m}{1 + \varepsilon_{m+1}},$$

and since $\varepsilon_m \uparrow 2$ if $m \uparrow \infty$, the lemma follows.

4. STABILITY IN THE MONOTONE CASE

For simplicity we begin with the case where $n = 2$ and D, D_h are polygonal domains as in §3, satisfying (3.1), and D is convex.

Set

$$(4.1) \quad f_h = u_h|_{\partial\Omega}, \quad f = u|_{\partial\Omega},$$

and let Γ be a nonempty open subset of $\partial\Omega$.

Theorem 4.1. *Under the foregoing assumptions*

$$(4.2) \quad \liminf_{h \rightarrow 0} \int_{\Gamma} \frac{|f_h - f|}{h} > 0.$$

This means that

$$(4.3) \quad d(D_h, D) \leq C \int_{\Gamma} |f_h - f|,$$

where the constant C may depend on the family $\{\sigma_h\}$. We note that the reverse inequality

$$(4.4) \quad \int_{\Gamma} |f_h - f| \leq Cd(D_h, D)$$

can easily be established.

From (3.9) we get, by integration by parts,

$$(4.5) \quad \begin{aligned} & - \int_{\Omega} a_h U_h \Delta v + (k-1) \int_{\partial D_h} U_h \frac{\partial v}{\partial \nu_e} + \int_{\partial \Omega} \frac{f_h - f}{h} \frac{\partial v}{\partial n} \\ & + \frac{k-1}{h} \int_{D_h \setminus D} \nabla u \cdot \nabla v = 0 \quad \forall v \in H^2(\Omega) \end{aligned}$$

where ν_e is the exterior normal to ∂D_h and n is the exterior normal to $\partial \Omega$.

Suppose (4.2) is not true, i.e., for a sequence $h \rightarrow 0$,

$$(4.6) \quad \int_{\Gamma} \frac{|f_h - f|}{h} \rightarrow 0.$$

Since $\Delta U_h = 0$ in $\Omega \setminus \overline{D_h}$ and in D_h , U_h is uniformly bounded in $L^2(\Omega)$ (by Lemma 3.1), we may assume that

$$(4.7) \quad U_h \rightarrow U \text{ uniformly in compact subsets of } \overline{\Omega} \setminus \partial D.$$

Since further $U = 0$ on Γ (by (4.6)) and $\partial U / \partial n = 0$ on $\partial \Omega$ we have, by unique continuation of harmonic functions,

$$(4.8) \quad U = 0 \quad \text{in } \Omega \setminus \overline{D};$$

also

$$(4.9) \quad \Delta U = 0 \quad \text{in } D.$$

Consequently

$$(4.10) \quad - \int_{\Omega} a_h U_h \Delta v \rightarrow - \int_D k U \Delta v \quad \text{as } h \rightarrow 0.$$

We next prove that

$$(4.11) \quad \int_{\partial D_h} U_h \frac{\partial v}{\partial \nu_e} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Since ∂D_h consists of N edges $\Gamma_j(h)$, it suffices to prove that

$$(4.12) \quad \int_{\Gamma_j(h)} U_h \frac{\partial v}{\partial \nu_e} \rightarrow 0$$

for each j .

Let σ_h be a line segment containing $\overline{\Omega_j(h)}$ in its interior and let

$$F_h(x) = \begin{cases} \partial v(x) / \partial \nu_e & \text{for } x \in \Gamma_j(h), \\ 0 & \text{for } x \in \sigma_h \setminus \Gamma_j(h). \end{cases}$$

Since F_h is piecewise smooth, it belongs to $W_p^{1-1/p}(\sigma_h)$ for any $p = 2 - \delta$, $\delta > 0$ [7, p. 45]. Hence, by the trace theorem [7, p. 37] there exists a function

z_h defined in semicircle S_h in $\Omega \setminus D_h$ with diameter σ_h such that

$$(4.13) \quad \|z_h\|_{W^{2,p}(S_h)} \leq C,$$

$$(4.14) \quad z_h = 0 \quad \text{on } \sigma_h,$$

$$(4.15) \quad \frac{\partial z_h}{\partial \nu_e} = F_h \quad \text{on } \sigma_h,$$

and

z_h vanishes in a neighborhood of $\partial S_h \setminus \sigma_h$.

It follows that

$$(4.16) \quad \int_{\Gamma_j(h)} U_h \frac{\partial v}{\partial \nu_e} = \int_{S_h} U_h \Delta z_h.$$

As $h \rightarrow 0$, $\Gamma_j(h) \rightarrow \Gamma_j$ and $\sigma_h \rightarrow \sigma$, $S_h \rightarrow S$. By regularity of v , $\partial v / \partial \nu_e$ on σ_h converges in $W_p^{1-1/p}$ -norm (when the independent variable is properly normalized so as to vary in the same interval σ , say). By the continuity of lifts (see [7, p. 37]) we then have that $z_h \rightarrow z$ in $W_{\text{loc}}^{2,p}(S)$. Recalling Lemma 3.1 we conclude that

$$\int_{S_h} U_h \Delta z_h \rightarrow \int_S U \Delta z,$$

and the right-hand side is equal to zero by (4.8). This completes the proof of (4.11).

Next we observe that, by (4.7),

$$(4.17) \quad \int_{\partial \Omega} \frac{f_h - f}{h} \frac{\partial v}{\partial n} = \int_{\partial \Omega} U_h \frac{\partial v}{\partial n} \rightarrow \int_{\partial \Omega} U \frac{\partial v}{\partial n} = 0.$$

We finally evaluate the last integral on the left-hand side of (4.5). Let T_j be the intersection of $D_h \setminus D$ with a square of side δ centered at the vertex s_j of ∂D . We can trace T_j by two families of intervals $l_{1j}(\lambda)$, $l_{2j}(\lambda)$, where the $l_{1j}(\lambda)$ are parallel to Γ_j at distance λ and the $l_{2j}(\lambda)$ are parallel to Γ_{j+1} at distance λ . Using (2.4) we get

$$(4.18) \quad \begin{aligned} & \left| \frac{k-1}{h} \int_{T_j} \nabla u \cdot \nabla v \right| \\ & \leq \frac{C}{h} \left[\int d\lambda \left\{ \int_{l_{1j}(\lambda)} |\nabla u|^2 \right\}^{1/2} + \int d\lambda \left\{ \int_{l_{2j}(\lambda)} |\nabla u|^2 \right\}^{1/2} \right] \\ & \leq \frac{C}{h} h = C\varepsilon(\delta) \rightarrow 0, \quad \varepsilon(\delta) \rightarrow 0 \text{ if } \delta \rightarrow 0. \end{aligned}$$

The set $D_h \setminus (D \cup (\bigcup T_j))$ is a disjoint union of rectangles $Q_{j,\delta}$ with two sides nearly parallel at distance $c(h)h$ ($\max c(h) = c_0 > 0$) and the other two sides lying near s_j and s_{j+1} . Since u is smooth in $\bar{\Omega} \setminus D$ except at the set of points $s_1, \dots, s_N$, we deduce that

$$\frac{1}{h} \int_{Q_{j,\delta}} \nabla u \cdot \nabla v \rightarrow \int_{\Gamma_{j,\delta}} \tilde{\sigma} \nabla u^e \cdot \nabla v, \quad \tilde{\sigma} \geq 0,$$

where $\Gamma_{j,\delta} \subset \Gamma_j$, and the right-hand side converges to

$$\int_{\Gamma_j} \tilde{\sigma} \nabla u^e \cdot \nabla v$$

as $\delta \rightarrow 0$. Combining this with (4.18) it follows that

$$(4.19) \quad \frac{k-1}{h} \int_{D_h \setminus D} \nabla u \cdot \nabla v \rightarrow (k-1) \int_{\partial D} \tilde{\sigma} \nabla u^e \cdot \nabla v \quad \forall v \in H^2(\Omega).$$

Notice that $\tilde{\sigma}$ is actually a linear function on each edge Γ_j , and

$$(4.20) \quad \tilde{\sigma} \geq 0, \tilde{\sigma} \not\equiv 0 \quad \text{on } \partial D.$$

We now take $h \rightarrow 0$ in (4.5) and use (4.19), (4.17), (4.11) and (4.10); we obtain

$$(4.21) \quad k \int_D U \Delta v = (k-1) \int_{\partial D} \tilde{\sigma} \nabla u^e \cdot \nabla v$$

for any $v \in H^2(\Omega)$.

Let V_ε be an ε -neighborhood of D . If $v \in H^2(V_\varepsilon)$ then we can modify it outside $V_{\varepsilon/2}$ so as to obtain a function $\tilde{v}$ in $H^2(\Omega)$. Since (4.21) is valid for $\tilde{v}$, it is also valid for v . Thus (4.21) holds for any $v \in H^2(V_\varepsilon)$.

The function $u^i = u|_D$ is smooth in D and therefore for any $x_0 \in D$ and $0 < \lambda < 1$, the function

$$(4.22) \quad v_\lambda(x) \equiv u^i(x_0 + \lambda(x - x_0)) \quad \text{is in } H^2(V_\varepsilon)$$

for some $\varepsilon > 0$. Substituting $v = v_\lambda$ into (4.21), we get

$$\int_{\partial D} \tilde{\sigma}(x) \nabla u^e(x) \cdot \nabla u^i(x_0 + \lambda(x - x_0)) = 0.$$

Letting $\lambda \uparrow 1$ and using Lemma 2.2, we easily conclude that

$$(4.23) \quad \int_{\partial D} \tilde{\sigma}(x) \nabla u^e(x) \cdot \nabla u^i(x) = 0.$$

Since finally

$$k \frac{\partial u^i}{\partial \nu_e} = \frac{\partial u^e}{\partial \nu_e}, \quad \frac{\partial u^i}{\partial \tau} = \frac{\partial u^e}{\partial \tau} \quad \text{on } \partial D$$

where τ is the tangential direction, it follows that

$$\int_{\partial D} \tilde{\sigma} |\nabla u^i|^2 = 0.$$

Recalling (4.20) we deduce that $\nabla u^i = 0$ on some arc on ∂D and hence, by harmonic continuation, $u = \text{const}$ in Ω . This implies, in particular, that $g = \partial u^e / \partial n \equiv 0$, which is a contradiction.

As we shall see below, Theorem 4.1 can be extended to general piecewise smooth domains D, D_h .

Definition 4.1. If each $\Gamma_j(h)$ is $C^{1,1}$ curve (instead of a line segment) with $C^{1,1}$ -norm bounded independently of h , and if the angles $\beta_j(h)$ at $s_j(h)$ satisfy

$$0 < c_1 \leq \beta_j(h) \leq c_2 < 2\pi \quad \forall j$$

then we say that D_h is uniformly piecewise $C^{1,1}$. Similarly we define “ D is piecewise $C^{1,1}$.”

We shall need the following assumptions:

(A₁) D_h are uniformly piecewise $C^{1,1}$ and D is piecewise $C^{1,1}$; further, D is strongly starshaped with the respect to the origin in the sense that $\bar{D} \subset \mu D$ for any $\mu > 1$.

(A₂) The vertices $s_j(h)$ of D_h and s_j of D are such that

$$|s_j(h) - s_j| \leq Ch.$$

(A₃) The representation (1.7) holds outside some $\delta(h)$ -neighborhood of $\{s_1, \dots, s_N\}$ where $\delta(h) \rightarrow 0$ if $h \rightarrow 0$; further,

$$\sigma_h(s) \rightarrow \sigma(s) \neq 0 \quad \text{as } h \rightarrow 0$$

uniformly outside any δ_0 -neighborhood of $\{s_1, \dots, s_N\}$.

(A₄) $D_h \supset D$ or $D_h \subset D \quad \forall 0 < h \leq h_0$.

Theorem 4.2. *Under the assumptions (A₁)–(A₄), the stability property (4.2) holds.*

The proof is similar to the proof of Theorem 4.1. The main difference occurs in the estimates near a vertex. Here we first perform a local diffeomorphism so as to make D locally a sector, and then proceed as before, with minor changes.

We now proceed to the case of dimension $n \geq 2$.

Definition 4.2. Let D be a domain in $\mathbb{R}^n$ ($n \geq 2$). We shall say that D is piecewise $C^{1,1}$ if for any $x_0 \in \partial D$ there exists a polyhedron D_* in $\mathbb{R}^n$, a point $x_* \in \partial D_*$ and a $C^{1,1}$ diffeomorphism G_{x_0} from a ball $B(x_0, \delta)$ onto a ball $B(x_*, \delta_*)$ such that

$$G_{x_0}(B(x_0, \delta) \cap D) = B(x_*, \delta_*) \cap D_*.$$

Definition 4.3. A family of domains D_h ($0 < h \leq h_0$) is said to be uniformly piecewise $C^{1,1}$ if in Definition 4.2 δ can be chosen independently of h , and the diffeomorphism $G_{x_0} \equiv G_{x_0, h}$ have $C^{1,1}$ norms bounded independently of h .

Let D be a piecewise $C^{1,1}$ bounded domain in $\mathbb{R}^n$ ($n \geq 2$), and let D_h be bounded domains in $\mathbb{R}^n$, uniformly piecewise $C^{1,1}$. Assume that ∂D_h is given by

$$(4.24) \quad \partial D_h: x = x_0 + h\sigma_h(x_0)\nu(x_0)$$

outside $\delta(h)$ -neighborhood of the set S of points of ∂D where ∂D is not $C^{1,1}$; here $\nu(x_0)$ is the outward normal, $\delta(h) \rightarrow 0$ if $h \rightarrow 0$, and

$$(4.25) \quad \begin{aligned} & |\sigma_h(x_0)| + |\nabla_{x_0}\sigma_h(x_0)| \leq C, \\ & \sigma_h(x_0) \rightarrow \sigma(x_0) \neq 0 \quad \text{as } h \rightarrow 0. \end{aligned}$$

Remark 4.1. As shown in [1], if the D_h are obtained from D by affine transformations, then (4.25) is valid.

Theorem 4.4. *Under the foregoing assumptions, if D is strongly star-shaped with respect to the origin and (A_4) is satisfied then the stability property (4.2) holds.*

The proof is similar to the proof of Theorems 4.1 and 4.2. In fact, once we can prove it for the case where D, D_h are polyhedra, the proof for the general case follows by using the same estimates after performing local diffeomorphism about points of the set S .

In proving the theorem for polyhedra D, D_h , the main new effort is in extending Lemma 2.2 (upon which Lemma 3.1 depends). Here we can probably again apply eigenfunction expansion to ζu where ζ is a cut-off function. We shall not attempt to carry it out since a proof of Lemma 2.2, which is valid in fact for any Lipschitz domain D (f is assumed to belong to $L^{2n/(n+1)}$), was recently given by Escuriaza and Fabes [2]. We note however that Lemma 2.1 (used in the proof of Lemma 2.2) will be needed in §6; it is mainly for this reason that we have included in this paper our original proofs of Lemmas 2.1 and 2.2.

We finally remark that if $n \geq 3$ we only need to use (3.8) for $\varepsilon = 0$. Indeed, for $n = 2$ (3.8) with $\varepsilon > 0$ was used only in establishing (4.11). In the present case of $n \geq 3$, the trace theorem [7, p. 37] allows $p = 2$ in (4.13), (4.14) and (4.15); thus (4.11) follows by using (3.8) with $\varepsilon = 0$.

Remark 4.2. The star-shaped assumption on D was used only in order to establish (4.22) for any $0 < \lambda < 1$. If D is in $C^{1,1}$ then the star-shaped assumption may be dropped since $u^i(x)$ is in $H^2(D)$ and can therefore be extended into a function in $H^2(V_\varepsilon)$.

5. THE NONMONOTONE CASE

From now on we drop the monotonicity assumption (1.6) but assume that

(5.1) there exists a diffeomorphism $y = x + \phi_h(x)$ of Ω onto Ω which maps D onto D_h and satisfies

$$|\nabla_x \phi_h(x)| \leq Ah \quad (A \text{ constant}).$$

Lemma 5.1. *Let D_h be uniformly piecewise $C^{1,1}$ domain and let D be such that (4.24) and (4.25) hold. Assume also that (5.1) is satisfied. If the stability property (4.2) is not satisfied then*

$$(5.2) \quad k \int_D U \Delta v = (k-1) \int_{\partial D} \tilde{\sigma} \nabla \tilde{u} \cdot \nabla v \quad \forall v \in H^2(\Omega)$$

holds, where

$$(5.3) \quad \Delta \tilde{u} = \begin{cases} \nabla u^e & \text{if } \tilde{\sigma}(x) > 0, \\ \nabla u^i & \text{if } \tilde{\sigma}(x) < 0, \end{cases}$$

and $\tilde{\sigma}(x)$ is a continuous function, $\tilde{\sigma}(x) \not\equiv 0$ and

$$(5.4) \quad \operatorname{sgn} \tilde{\sigma}(x) = \operatorname{sgn} \sigma(x).$$

The proof is similar to the proof of (4.21) for polygonal domains in the monotone case; for $C^{1,1}$ domain the theorem was already proved in [1]. The main difference in the proof for the piecewise $C^{1,1}$ case occurs in establishing (4.19); it is here that the assumption (5.1) is needed (cf. [1], following the proof of Lemma 3.3).

Corollary 5.2. *If $\sigma(x) = 0$ on a nonempty open subset of $\partial\Omega$ then the stability property (4.2) holds.*

Indeed, this follows from the proof of Lemma 5.1 in precisely the same way as Corollary 3.4 of [1] which dealt with the case where D and D_h are $C^{1,1}$ domains.

In the remaining part of this section we assume that

$$(5.5) \quad n = 2 \quad \text{and} \quad \partial D \text{ is analytic.}$$

This implies that u^e is analytic on ∂D . We shall prove that, for appropriately chosen g , the stability property (4.2) holds.

In addition to (5.5) we shall assume that

$$(5.6) \quad \partial D \text{ is strongly star-shaped with respect to the origin,}$$

and that

$$(5.7) \quad \sigma(x) \text{ changes sign along } \partial D \text{ only a finite number of times.}$$

The assumption (5.6) is made so that one may apply (5.2) to a function as in (4.22) ($0 < \lambda < 1$) and thus deduce, as $\lambda \uparrow 1$, that if the stability property (4.2) does not hold then

$$(5.8) \quad \int_{\partial D} \tilde{\sigma} \nabla \tilde{u} \cdot \nabla v = 0 \quad \text{if } \Delta v = 0 \text{ in } D \text{ and } \nabla v \in L^1(\partial D).$$

One can actually easily verify the condition (5.1) when (5.5) and (5.6) hold.

Assumption (5.7) implies that

$$(5.9) \quad \begin{aligned} \{\tilde{\sigma} > 0\} &= \bigcup_{j=1}^M I_j^+, \\ \{\tilde{\sigma} < 0\} &= \bigcup_{j=1}^{M'} I_j^- \quad \text{where } I_j^+, I_k^- \text{ are disjoint arcs on } \partial D. \end{aligned}$$

By (5.3),

$$(5.10) \quad \nabla \tilde{u} = \begin{cases} \nabla u^e & \text{on } \bigcup_{j=1}^M I_j^+, \\ \nabla u^i & \text{on } \bigcup_{j=1}^{M'} I_j^-. \end{cases}$$

In view of Corollary 5.2, we may assume from now on that $M' = M$ and the union of the $\overline{I_j^+}, \overline{I_k^-}$ is all of ∂D .

Lemma 5.3. *If for any C^1 function h there exists a solution w to*

$$(5.11) \quad \begin{aligned} \Delta w &= 0 \quad \text{in } D, \\ \nabla \tilde{u} \cdot \nabla w &= h \quad \text{on } \partial D, \\ \nabla w &\in L^1(\partial D), \end{aligned}$$

then the stability property (4.2) holds.

Indeed, using (5.8) we conclude that

$$\int_{\partial D} \tilde{\sigma} h = 0;$$

since h is arbitrary, $\tilde{\sigma} = 0$ which is a contradiction.

Remark 5.1. It is actually sufficient to solve (5.11) just for $h = \tilde{\sigma}$, but $\tilde{\sigma}$ may not be C^1 .

In order to establish (5.11) we shall rely on the index theory for the Riemann-Hilbert problem as exposed in [8].

We recall (see [8, §40, (40.8)]) that for a continuous vector field $V = a + ib$ on $\overline{\Omega}$ and a smooth curve $\Gamma \subset \overline{\Omega}$ which is the boundary of a subdomain in Ω one defines the index of V with respect to Γ by

$$(5.12) \quad \kappa(V; \Gamma) = \frac{1}{\pi} [\arg(a - ib)]_{\Gamma}$$

provided $V \neq 0$ on Γ .

It is well known that the index is homotopic invariant, i.e., if $V(\theta)$ is a family of such vector fields continuous in θ , $0 \leq \theta \leq 1$, then

$$(5.13) \quad \kappa(V(0); \Gamma) = \kappa(V(1); \Gamma)$$

provided $V(\theta) \neq 0$ on Γ for all θ . The definition of the index of $V = a + ib$ with respect to Γ can be extended to the case where V may vanish or have finite number of jump-discontinuities at points $c_1, \dots, c_N$ on Γ . Setting $G = (a - ib)/(a + ib)$ one defines (see [8, §93, p. 273]),

$$(5.14) \quad \kappa(V; \Gamma) = \frac{1}{2\pi} [\arg G]_{\Gamma}$$

provided the limits $G(c_i \pm 0)$ exist, where the passage from $G(c_i + 0)$ to $G(c_i - 0)$ is selected as [8, §85]. If $V(\theta)$ varies continuously with θ and each $V(\theta)$ vanishes or has jump-discontinuities only at $c_1, \dots, c_N$ then (5.13) is still valid provided the limits $G(c_i \pm 0)$ exist for all $0 \leq \theta \leq 1$.

Consider the example of

$$(5.15) \quad V_0(\theta) = \left(1 - \frac{(k-1)\theta}{k}\right) u_N^e \vec{N} + u_t^e \vec{\tau} \quad (0 \leq \theta \leq 1)$$

where $\vec{N}$ is the outward unit normal and $\vec{\tau}$ is unit tangent (in the counter-clockwise direction) along ∂D . From the diffraction conditions

$$(5.16) \quad u^e = u^i, \quad \frac{\partial u^e}{\partial N} = k \frac{\partial u^i}{\partial N}$$

we see that, for any $z \in \partial D$,

$$\nabla u^e(z) \neq 0 \quad \text{if and only if} \quad \nabla u^i(z) \neq 0.$$

Since $V_0(\theta)$ is a homotopy from ∇u^e to ∇u^i , we conclude that

$$\kappa(\nabla u^e; \partial D) = \kappa(\nabla u^i; \partial D) \quad \text{if } \nabla u^e \neq 0 \text{ on } \partial D.$$

Consider next the vector field

$$V_1(\theta) = \begin{cases} V_0(\theta) & \text{on } \partial D^+ \equiv \bigcup_j I_j^+, \\ \nabla u^i & \text{on } \partial D^- \equiv \bigcup_j I_j^-. \end{cases}$$

Clearly $V_1(0) = \nabla \tilde{u}$ and $V_1(1) = \nabla u^i$. Notice that $V_1(\theta)$ has a finite number of jump discontinuities along ∂D , i.e., at the endpoints of the I_j^- . As explained above the invariance formula (5.13) is still valid, so that

$$(5.18) \quad \kappa(\nabla u^i; \partial D) = \kappa(\nabla \tilde{u}; \partial D) \quad \text{provided } \nabla u^i \neq 0 \text{ on } \partial D.$$

The Riemann-Hilbert problem in D is concerned with finding a holomorphic function ϕ in D , continuous in $\overline{D}$, such that

$$a \operatorname{Re} \phi + b \operatorname{Im} \phi = c \quad \text{on } \partial D;$$

here D is a C^1 domain and a, b, c are piecewise continuous with a finite number of discontinuities $z_1, \dots, z_N$, and their derivatives are bounded in each arc $z_j z_{j+1}$ ($z_{N+1} = z_1$). By [8, §93], if $a^2 + b^2 > 0$ and the index of $V = a + ib$ with respect to ∂D is ≥ -1 then for any c there exists a solution ϕ and [8, (93.1)],

$$|\phi(z)| \leq \frac{C}{|z - z_j|^\alpha} \quad (0 < \alpha < 1)$$

for z near z_j .

We note that the Riemann-Hilbert problem for holomorphic function $\phi(z)$ is equivalent to the problem

$$\begin{aligned} \Delta v &= 0 \quad \text{in } D, \\ av_x + bv_y &= c \quad \text{on } \partial D, \end{aligned}$$

for $v = \operatorname{Re} \int \phi(z) dz$. We therefore conclude:

Lemma 5.4. *If $\kappa(\nabla \bar{u}; \partial D) \geq -1$ then for any piecewise C^1 function h there exists a solution to (5.11).*

The fact that $\nabla w \in L^1(\partial D)$ follows from the estimate of $\phi(z)$ near z_j , where z_j are the points of discontinuity of V . Recalling (5.18) we have thus reduced the proof of the stability property (4.2) to showing that

$$(5.19) \quad \nabla u^i \neq 0 \quad \text{on } \partial D$$

and

$$(5.20) \quad \kappa(\nabla u^i; \partial D) \geq -1.$$

Since u^i is analytic in $\overline{D}$, it has analytic extension into a neighborhood N^+ of $\overline{D}$; we denote it by u^i and note that u^i is harmonic in N^+ . Similarly u^e has analytic (and harmonic) extension into an Ω -neighborhood N^- of $\Omega \setminus D$.

We shall now make a special choice of g as follows:

Definition of g . Let $z = z(t)$ be a $C^{1,\alpha}$ parametrization of $\partial \Omega$ ($0 \leq t \leq 2\pi$) and let $f(z(t))$ be a $C^{1,\alpha}$ function such that $f(z(t))$ has a unique maximum at $t = 0$, a unique minimum at some point $t = t_0$, and

$$\begin{aligned} \frac{d}{dt} f(z(t)) &< 0 \quad \text{if } 0 < t < t_0, \\ \frac{d}{dt} f(z(t)) &> 0 \quad \text{if } t_0 < t < 2\pi. \end{aligned}$$

Let u be the solution of the diffraction problem

$$(5.21) \quad \operatorname{div}(a \nabla u) = 0 \quad \text{in } \Omega,$$

$$(5.22) \quad u = f \quad \text{on } \partial \Omega,$$

and set

$$(5.23) \quad g = \frac{\partial u}{\partial \nu}.$$

We shall prove

Lemma 5.5. *For the special choice of g in (5.23), (5.19) holds and*

$$(5.24) \quad \kappa(\nabla u^i; \partial D) = 0.$$

Since (5.24) implies (5.20), we deduce

Theorem 5.6. *Under the assumptions (5.5)–(5.7), the stability property (4.2) holds.*

Proof of Lemma 5.5. From the transmission conditions (5.16) one can easily show that u cannot take minimum or maximum at points on ∂D . Therefore u attains its maximum at $z(0)$ and its minimum at $z(t_0)$ and, by the maximum principle, $\partial u / \partial \nu \neq 0$ at these two points. At all other points of $\partial \Omega$ we also have $\partial u / \partial \tau \neq 0$ (by the choice of f). Consequently $\nabla u^e \neq 0$ on $\partial \Omega$. Since further the tangential components of ∇u^e have the same sign on $(0, t_0)$ and (the reverse sign) on $(t_0, 2\pi)$, it can be seen that

$$(5.25) \quad \kappa(\nabla u^e; \partial \Omega) = 0.$$

The vector field ∇u^i has a finite number of zeros $z_1, \dots, z_m$ in $\bar{D}$ and similarly (since $\nabla u^e \neq 0$ in an Ω -neighborhood of $\partial \Omega$) the vector field ∇u^e has a finite number of zeros $z_{m+1}, \dots, z_\sigma$ in $\Omega \setminus D$. On ∂D , ∇u^i and ∇u^e have common zeros (if any); we denote them by $z_{l+1}, z_{l+2}, \dots, z_m$.

Let L_ε be the Jordan curve formed by the arcs of the $\partial B(z_j; \varepsilon)$ ($l+1 \leq j \leq m$) which are contained in $\Omega \setminus D$ and by $\partial D \setminus \bigcup_{j=l+1}^m B(z_j; \varepsilon)$. We claim

Lemma 5.7. *If ε is sufficiently small then*

$$(5.26) \quad \nabla u^e|_{L_\varepsilon} \text{ is homotopic to } \nabla u^i|_{L_\varepsilon}.$$

Proof. If ε is sufficiently small then $\nabla u \neq 0$ in a neighborhood of

$$\Gamma_\varepsilon \equiv \partial D \setminus \bigcup_{l+1}^m B(z_j; \varepsilon).$$

Let $\nabla u^e = u_N^e \vec{N} + u_\tau^e \vec{\tau}$. Then u_N^e and u_τ^e do not vanish simultaneously on Γ_ε . We define, for $0 \leq \theta \leq 1$,

$$(5.27) \quad V(\theta) = \left(1 - \frac{k-1}{k}\theta\right) u_N^e \vec{N} + u_\tau^e \vec{\tau} \quad \text{on } \Gamma_\varepsilon.$$

Then $V(0) = \nabla u^e$, $V(1) = \nabla u^i$ and $V(\theta)$ is continuous in θ ; moreover, $V(\theta) \neq 0$ on $\bar{\Gamma}_\varepsilon$.

We next wish to define $V(\theta)$ on any arc $\partial B(z_j; \varepsilon) \setminus \bar{D}$ of L_ε . To do this we introduce a conformal mapping of the lower half-plane onto D which maps 0 into z_j . By analytic continuation, the mapping is conformal in a neighborhood of 0. Since the refraction conditions (5.16) are invariant under conformal mapping, we may assume from the start that $z = 0$ and that D , near $z = 0$, coincides with the lower half-plane. Expanding u^e , u^i near $z = 0$ into series

$$\begin{aligned} u^e &= \sum r^n (a_n^e \cos n\varphi + b_n^e \sin n\varphi), \\ u^i &= \sum r^n (a_n^i \cos n\varphi + b_n^i \sin n\varphi) \end{aligned}$$

and using the refraction conditions, we obtain

$$(5.28) \quad a_n^e = a_n^i, \quad b_n^e = k b_n^i.$$

We now define (in the variables (r, φ) of the conformal mapping) $V(\theta) = \nabla u(\theta)$ where

$$u(\theta) = \sum r^n \left(a_n^e \cos n\varphi + \left(1 - \frac{k-1}{k} \theta \right) b_n^e \sin n\varphi \right).$$

Then, by (5.28), $u(0) = u^e$, $u(1) = u^i$ so that

$$V(0) = \nabla u^e, \quad V(1) = \nabla u^i.$$

Further,

$$\nabla V(\theta) \neq 0 \quad \text{on } \partial B(z_j; \varepsilon)$$

and $V(\theta)$ continuously fits with (5.27) at the two points of $\partial \Gamma_\varepsilon \cap \partial B(z_j; \varepsilon)$.

We have thus constructed a homotopy $V(\theta)$ of ∇u^i along L_ε ; this establishes the assertion (5.26).

Completion of the proof of Lemma 5.5. Consider the index

$$\kappa(\zeta) \equiv \kappa(\nabla u; \partial B(\zeta; \varepsilon)) \quad (\varepsilon \text{ small})$$

of ∇u at a zero $k = \zeta$ of ∇u , where $\zeta \in \Omega \setminus \partial D$. Introducing $h = u + iv$ (h holomorphic), we have

$$h'(z) = u_x + iv_x = u_x - iu_y,$$

and

$$\begin{aligned} \pi \kappa(\zeta) &= \pi \kappa(u_x - iu_y; \partial B(\zeta; \varepsilon)) = \text{Var}_{\partial B(\zeta; \varepsilon)} h'(z) \\ &= \text{Var}_{\partial B(\zeta; \varepsilon)} (z - \zeta)^{n-1} = 2\pi(n-1) \end{aligned}$$

where $h(z) = a_0(z - \zeta)^n + \dots$, $a_0 \neq 0$. Hence

$$(5.29) \quad \kappa(\zeta) = 2 \times \{\text{order of zero of } \nabla u \text{ at } z = \zeta\}.$$

Denote by π_j the order of the zero of ∇u at $z = z_j$. Then, for small $\varepsilon > 0$,

$$(5.30) \quad \kappa(\nabla u^e; \partial \Omega) = \kappa(\nabla u^e; L_\varepsilon) + \sum_{j=m+1}^{\sigma} 2\pi_j.$$

From Lemma 5.7 we also have

$$(5.31) \quad \kappa(\nabla u^e; L_\varepsilon) = \kappa(\nabla u^i; L_\varepsilon).$$

Finally, by (5.29),

$$(5.32) \quad \kappa(\nabla u^i; L_\varepsilon) = \sum_{j=1}^m 2\pi_j.$$

Combining (5.30)–(5.32) and recalling (5.25) we deduce that

$$\sum_{j=1}^{\sigma} \pi_j = 0.$$

Hence ∇u has no zeros in $\overline{D}$ and (5.24) holds. This completes the proof of Lemma 5.5 and therefore also of Theorem 5.6.

6. THE NONMONOTONE CASE WITH PIECEWISE ANALYTIC ∂D

In this section we continue to consider the nonmonotone case for $n = 2$, but assume that ∂D is piecewise analytic with a finite number of vertices

$s_1, \dots, s_N$. For simplicity we take D and D_h to be polygons, as in §2, with D convex, and

$$c_1 h \leq \sum_{j=1}^N |s_j(h) - s_j| \leq c_2 h \quad (0 < c_1 < c_2 < \infty);$$

however our results easily extend to any piecewise analytic ∂D .

Our starting point is Lemma 5.1; as in the proof of Theorem 5.6 (recall Remark 5.1) the stability property (4.2) holds if there exists a solution v to

$$(6.1) \quad \Delta v = 0 \quad \text{in } D,$$

$$(6.2) \quad \nabla \tilde{u} \cdot \nabla v = \tilde{\sigma} \quad \text{on } \partial D,$$

such that $\nabla v_\lambda \rightarrow \nabla v$ in $L^1(\partial D)$ as $\lambda \uparrow 1$; $v_\lambda(x) = v(x_0 + \lambda(x - x_0))$ for some $x_0 \in D$. The function $\tilde{\sigma}$ is linear on each edge Γ_j .

By Lemma 2.1 it follows that any $\nabla \tilde{u}$ has a finite variation along any arc of ∂D which contains a vertex s_j of ∂D_i ; consequently

$$(6.3) \quad \kappa(\nabla \tilde{u}; \partial D) < \infty.$$

In the original theory of Muskhelishvili [8, §§93, 94] the domain D is assumed to be smooth. However, the results remain true if ∂D is piecewise C^1 ; see [3, Example 8.4]. One can see it by using conformal mapping $z = z(\omega)$ of the unit disc $\{|\omega| < 1\}$ onto D , and applying the original theory in the ω -domain noting that the index of $z'(\omega)$ is zero since $\omega \rightarrow z(\omega)$ is conformal (i.e. $z'(\omega)$ does not vanish in the unit disc).

If $\kappa \geq -1$ then (by [8, §93]) there exists a unique solution of (6.1), (6.2) satisfying

$$(6.4) \quad |\nabla v| \leq \frac{C}{|x - c_j|^\alpha}.$$

On the other hand, if $\kappa \leq -2$ then there are $-\kappa - 1$ solutions of the homogeneous problem, and the solution v of (6.1), (6.2), (6.4) exists if and only if $\tilde{\sigma}$ is orthogonal to these solutions. The orthogonality relations can be written in the form

$$(6.5) \quad \int_{\partial D} \tilde{\sigma} l_m = 0, \quad m = 1, \dots, -\kappa - 1.$$

We summarize

Theorem 6.1. (i) If $\kappa \geq -1$ then the stability property (4.2) holds;
(ii) if $\kappa \leq -2$ and

$$(6.6) \quad \tilde{\sigma} \text{ is not a linear combination of } l_1, \dots, l_{-\kappa-1}$$

then the stability property (4.2) holds.

This result for analytic ∂D was proved in [1].

We now wish to estimate the index κ for the special choice of g made in §5.

From the results of [9, p. 201] it follows that the solution u of the diffraction problem (3.2) cannot take local maximum (or local minimum) at a vertex of ∂D . So as in the proof of (5.25):

Lemma 6.2. *For the special choice of g in (5.21)–(5.23),*

$$(6.7) \quad \kappa(\nabla u^e; \partial\Omega) = 0.$$

From now on we shall work with the special choice of g in (5.21)–(5.23).

Lemma 6.3. $\nabla u(x) \neq 0$ for $x \neq s_1, \dots, s_N$.

Proof. Let ξ_j, η_j be points on ∂D , $\xi_j \in \Gamma_j$ and $\eta_j \in \Gamma_{j+1}$, such that $|\xi_j - s_j|$ and $|\eta_j - s_j|$ are small and $\nabla u^e(\xi_j) \neq 0$, $\nabla u^e(\eta_j) \neq 0$. Let $\tilde{\Gamma}_\varepsilon$ be C^2 and piecewise analytic curves in $\bar{D}$ which converge to ∂D as $\varepsilon \rightarrow 0$ such that $\tilde{\Gamma}_\varepsilon$ connects ξ_j to η_j by an analytic arc and it is a line-segment between η_j and ξ_{j+1} (this segment lies on ∂D). Denote by $\tilde{D}_\varepsilon$ the domain bounded by $\tilde{\Gamma}_\varepsilon$, and let $\tilde{u}_\varepsilon$ denote the solution of the refraction problem corresponding to $\tilde{\Gamma}_\varepsilon$. Then

$$(6.8) \quad \nabla \tilde{u}_\varepsilon^i \rightarrow \nabla u^i \text{ and } \nabla \tilde{u}_\varepsilon^e \rightarrow \nabla u^e \text{ uniformly outside any neighborhood of } \{s_1, \dots, s_N\}.$$

Observe that $\nabla \tilde{u}_\varepsilon \neq 0$ at ξ_j, η_j for all ε , and that $\nabla \tilde{u}_\varepsilon^e, \nabla \tilde{u}_\varepsilon^i$ are analytic across $\tilde{\Gamma}_\varepsilon \setminus \{\xi_1, \eta_1, \dots, \xi_N, \eta_N\}$. Therefore $\nabla \tilde{u}_\varepsilon^e$ and $\nabla \tilde{u}_\varepsilon^i$ have only a finite number of zeros. Since also $\kappa(\nabla \tilde{u}_\varepsilon; \partial\Omega) = 0$, we can repeat an argument used in §5 and deduce that

$$(6.9) \quad \begin{aligned} \nabla \tilde{u}_\varepsilon^e(x) &\neq 0 \text{ in } \bar{\Omega} \setminus \tilde{\Gamma}_\varepsilon, \\ \nabla \tilde{u}_\varepsilon^i(x) &\neq 0 \text{ in } \tilde{D}_\varepsilon \cup \tilde{\Gamma}_\varepsilon. \end{aligned}$$

We now suppose that $\nabla u^e(x_0) = 0$ for some $x_0 \in \Omega \setminus D$, $x_0 \neq \text{vertex}$. If $x_0 \in \partial D$ then we choose the ξ_j, η_j above so that x_0 lies on one of the line segments of $\tilde{\Gamma}_\varepsilon$. Then (whether $x_0 \in \partial D$ or $x_0 \notin \partial D$) there exists a small disc $B_\delta(x_0)$ such that u^e and $\tilde{u}_\varepsilon^e$ are analytic in $\bar{B}_\delta(x_0)$ and $\nabla u^e \neq 0$ on $\partial B_\delta(x_0)$. From (6.8), (6.9) we then deduce that

$$0 = \kappa(\nabla \tilde{u}_\varepsilon^e; x_0) = \kappa(\nabla \tilde{u}_\varepsilon^e; \partial B_\delta(x_0)) = \kappa(\nabla u^e; \partial B_\delta(x_0)) = \kappa(\nabla u^e; x_0)$$

if ε is small enough, which is a contradiction.

Similarly one can prove that $\nabla u^i(x) \neq 0$ if $x \in \bar{D}$, $x \neq \text{vertex}$.

Denote by $\tilde{\Gamma}_\varepsilon^e$ smooth curves in $\Omega \setminus D$ such that $\tilde{\Gamma}_\varepsilon^e$ coincides with ∂D outside ε -neighborhood of each s_j , and such that $\tilde{\Gamma}_\varepsilon^e$ connects a point in Γ_j to a point in Γ_{j+1} by an arc $\sigma_{j,\varepsilon}^e$ which “approximately” lies on $|z - s_j| = \varepsilon$; by “approximately” we mean that we make $\tilde{\Gamma}_\varepsilon^e$ smooth as it intersects ∂D , by slightly modifying the arc $|z - s_j| = \varepsilon$. Similarly we define curves $\tilde{\Gamma}_\varepsilon^i$ and the approximate arcs $\sigma_{j,\varepsilon}^i$ in D .

By Lemma 2.1, near s_j ,

$$(6.10) \quad \begin{aligned} \overline{\nabla u^e} &= A_j^e(z - s_j)^{\gamma_j - 1}(1 + o(1)), \\ \overline{\nabla u^i} &= A_j^i(z - s_j)^{\gamma_j - 1}(1 + o(1)), \end{aligned}$$

where $A_j^e \neq 0$, $A_j^i \neq 0$ and γ_j is a positive number larger than $1/2$. Indeed, take for simplicity $j = 1$, $s_j = 0$. By (2.21),

$$(6.11) \quad u = \operatorname{Re} \sum_{m=1}^{k-1} B_m z^{\gamma_m} + \sum_{m=k}^{\infty} C_m r^{\gamma_m} v_m(\theta), \quad B_1 \neq 0;$$

k is chosen so that $k > 1$ and $\gamma_k > 3 + \gamma_1$. By (2.13) and (2.11)

$$\int_0^{2\pi} (v'_m(\theta))^2 \leq C\gamma_m^2,$$

so that

$$(6.12) \quad |v_m(\theta)| \leq C\gamma_m.$$

Therefore, by (2.11),

$$|v''_m(\theta)| \leq C\gamma_m^3 \quad \text{if } \theta \neq 0, \theta \neq \beta$$

and then also

$$(6.13) \quad |v'_m(\theta)| \leq C\gamma_m^3.$$

Using (6.12), (6.13) and (2.22), (2.7), we deduce from (6.11) that

$$\overline{\nabla u} = \operatorname{Re} \sum_{m=1}^{k-1} \gamma_m B_m z^{\gamma_m-1} + O(r^{\gamma_k-3}),$$

which implies (6.10).

From Lemma 6.4 and Corollary 6.3 we have

$$(6.14) \quad \kappa(\nabla u^e; \tilde{\Gamma}_e^e) = 0,$$

$$(6.15) \quad \kappa(\nabla u^i; \tilde{\Gamma}_e^i) = 0.$$

Take any vertex s_j and introduce polar coordinates (r, θ) about s_j . We want to evaluate the index of ∇u^i with respect to the boundary of a small domain $\Omega_j \subset D$ such that $\partial\Omega_j$ consists of two line segments

$$l_1 = \{(r, 0), 0 \leq r \leq \delta_0\}, \quad l_2 = \{(r, \beta_j), 0 \leq r \leq \delta_0\}$$

and a circular arc

$$l_3 = \{(\delta_0, \theta), 0 \leq \theta \leq \beta_j\}.$$

Set $\Sigma_e = l_2 \cup l_1 \cup l_3$. By the definition of index in (5.14) and the expansion (6.10) for ∇u^i ,

$$\begin{aligned} \kappa(\nabla u^i; \Sigma_e) &= \frac{1}{2\pi} \arg \left[\frac{\overline{\nabla u^i}}{\nabla u^i} \right]_{l_3} + \frac{1}{2\pi} \arg \left[\frac{\overline{\nabla u^i}}{\nabla u^i} \right]_{l_2 \cup l_1} \\ &= \frac{(\gamma_j - 1)\beta_j}{\pi} + \frac{1}{\pi} \{\text{jump of } \arg(z - s_j)^{\gamma_j-1} \\ &\quad \text{from } \theta = \beta_j \text{ to } \theta = 0 \text{ at } r = 0\} \\ &= \frac{(\gamma_j - 1)\beta_j}{\pi} - \left\{ \text{nonintegral part of } \frac{(\gamma_j - 1)\beta_j}{\pi} \right\} \\ &= \left[\frac{(\gamma_j - 1)\beta_j}{\pi} \right] \end{aligned} \quad (6.16)$$

where $[x]$ is the integral part of x if $x > 0$ and $[x] = 0$ if $x < 0$ (if $\gamma_j - 1 < 0$ then, since $\gamma_j > \frac{1}{2}$, the right-hand side of (6.16) is nonpositive and smaller than 1 in absolute value; since the index is an integer, it must then be equal to zero).

Repeating the above argument at each vertex s_j and recalling (6.15) we deduce

Lemma 6.5. $\kappa(\nabla u^i; \partial D) \geq 0$.

To compute $\kappa(\nabla \tilde{u}; \partial D)$ we deform ∇u^i into $\nabla \tilde{u}$. Consider first the case where $\nabla \tilde{u}$ is obtained by replacing ∇u^i by ∇u^e on a single closed arc σ . If σ lies inside one edge Γ_j then

$$\kappa(\nabla u^i; \partial D) - \kappa(\nabla \tilde{u}; \partial D) = \frac{\lambda_1}{\pi} + \frac{\lambda_2}{\pi}$$

where λ_k is the difference in the arguments of ∇u^i and ∇u^e at an endpoint of σ . Since the index is an integer whereas $|\lambda_k| < \pi/2$, it follows that

$$(6.17) \quad \kappa(\nabla u^i; \partial D) - \kappa(\nabla \tilde{u}; \partial D) = 0.$$

Suppose next that σ contains a vertex s_j and its endpoints lie inside Γ_j and Γ_{j+1} . Then, by (5.14),

$$(6.18) \quad \kappa(\nabla u^i; \partial D) - \kappa(\nabla \tilde{u}; \partial D) = \frac{\lambda_1}{\pi} + \frac{\lambda_2}{\pi} + \frac{\Delta}{2\pi}$$

where Δ is the difference between

$$B^i e^{2\sqrt{-1}(\gamma_j-1)\beta_j} - B^i e^{2\sqrt{-1}(\gamma_j-1)0} \quad \left(B^i = \frac{\alpha_i - \sqrt{-1}\beta_i}{\alpha_i + \sqrt{-1}\beta_i}, \alpha_i + \sqrt{-1}\beta_i = A_j^i \right)$$

and the corresponding expression with B^e (associated with A_j^e). Since the index is an integer and $|(\lambda_1 + \lambda_2)/\pi| < 1$, there is no ambiguity about the choice of the corresponding limits $G(s_j \pm 0)$ if $|A^i - A^e|$ is small; in fact, the correct choice is such that the limits of

$$\frac{1}{B^i} \frac{\overline{\nabla u^i}}{\nabla u^i} \text{ and } \frac{1}{B^e} \frac{\overline{\nabla u^e}}{\nabla u^e} \text{ agree at } s_j \pm 0$$

so that the left-hand side of (6.18) is equal to zero. By continuously deforming A^e we deduce that (6.17) holds in the general case.

Finally, a similar argument shows that (6.17) holds if σ lies in Γ_j and one of its endpoints is a vertex of ∂D .

By deforming ∇u^i step-by-step a finite number of times so as to obtain $\nabla \tilde{u}$, and applying (6.17) at each step, we deduce that (6.17) is valid for general $\nabla \tilde{u}$. Consequently, by Lemma 6.5,

$$(6.19) \quad \kappa(\nabla \tilde{u}; \partial D) \geq 0.$$

This together with Theorem 6.1 implies

Theorem 6.6. *For the special choice of g made in (5.21)–(5.23), the stability property (4.2) holds.*

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